

Manuel Baltieri, Ph.D.

Agent foundations, world models, ALife, and safe AI systems

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PROFILE

- Research scientist working on agent foundations, world models, abstractions, and safe AI systems, using applied category theory, control theory, Bayesian inference, computational mechanics, and reinforcement learning.
- ARIA Creator, Co-PI on *Mathematics for Safe AI*, with recent work on agent-environment world models, compositional semantics for state abstractions in reinforcement learning and agent sandboxing.
- Research engineering in Python/PyTorch for state-space models, Bayesian filtering/control, simulation, LLM fine-tuning/evaluation, and program-synthesis prototypes, with earlier JAX/NumPyro porting for active inference models.

TECHNICAL STACK

Bayesian inference, state-space models, optimal control, reinforcement learning, LLM fine-tuning/evaluation, simulation, C/C++, MATLAB, Java, Cobol. Python, PyTorch, JAX/NumPyro for research prototyping and porting.

RESEARCH & ENGINEERING

Research Team Leader | ARIA R&D Creator (prev. Researcher) Nov 2021 - present

Araya Inc., Tokyo, JP

- **Agent foundations and Safe AI:** Lead research on mathematical frameworks for safe AI, working on world models, abstractions and agent-environment boundaries, using applied category theory, control theory, computational mechanics, Bayesian inference, and reinforcement learning. Authored work at ICML, RLC and ALIFE. ARIA R&D Creator, Co-PI for project on *Mathematics for Safe AI*.
- **LLM and program-synthesis engineering:** Led client-facing research engineering on PyTorch LLM program-synthesis models, including architecture optimisation, fine-tuning, prompt design, and evaluation, reducing parameter budget by 20% while preserving task performance, supporting contract renewal.
- **Human decision modelling:** Developed NeuroAI models of decision-making under uncertainty for a major automotive manufacturer, informing the client's experimental design and supporting contract renewal.

Research Consultant and Developer

Sep 2021 - Nov 2021

Nested Minds Network Ltd., USA

- **JAX/NumPyro research engineering:** Worked on porting of a PyTorch library for continuous-time state-space models, Bayesian filtering/control, and active inference to NumPyro/JAX, used across more than 10 internal projects.

JSPS/Royal Society Postdoctoral Research Fellow

Nov 2019 - Oct 2021

Laboratory for Neural Computation, RIKEN Center for Brain Science, JP

- **Bayesian inference / control:** Developed *Generalised Filtering*, a PyTorch package for continuous-state, continuous-time state-space models, Bayesian filtering, learning, and optimal control, and published 8 papers across AI/ML, artificial life, control theory, and neuroscience.

AWARDS AND GRANTS

Grants

- ARIA (Co-PI) - *Mathematics for Safe AI* (USD 380,000) 2025 - 2027
- Kakenhi (PI) - Grant-in-Aid for Scientific Research (USD 22,000) 2019 - 2021

Fellowships and scholarships

- JSPS/Royal Society Postdoctoral Fellowship (USD 80,000/2 years) 2019 - 2021
- oLife Postdoctoral Fellowship, Origins of Life Programme, NL (USD 140,000/3 years, declined to accept JSPS) 2019 - 2022

RESEARCH LEADERSHIP

- Board of Directors, International Society for Artificial Life Nov 2023 - present
- Programme Chair / Proceedings, ALIFE 2025, Proceedings and Local Chair, ALIFE 2023, Research Organiser, Mathematical Boundaries Workshop, 2023-2025.
- Co-organiser, *Goal-Directed Behaviour in Life and Non-Life*, ALIFE 2024, Lead Organiser, *Hybrid Life* special sessions, 2018-2024.
- Reviewer/meta-reviewer for Nature Communications, PNAS, IROS, ALIFE, ICLR workshops, and related venues.

SELECTED RESEARCH OUTPUTS

- Zhang, Y.*, Luo, Z., & **Baltieri, M.** (2026). Compositional Behavioral Semantics for State Abstraction in Reinforcement Learning. In *International Conference on Machine Learning - ICML 2026*.
- **Baltieri, M.***, & Suzuki, K. (2026). Mathematical approaches to the study of agents. *Philosophical Transactions of the Royal Society B*. To appear.
- **Baltieri, M.***, Torresan, F., Zhang, Y., Boyd, A., & Rosas, F. E. (2026). World models: environment, agent and joint agent-environment models. Submitted.
- Boyd, A.*, Nowak, F., Hyland, D., **Baltieri, M.**, & Rosas, F. E. (2025). From monoliths to modules: decomposing transducers for efficient world modelling. *arXiv preprint arXiv:2512.02193*.
- Torresan, F., Suzuki, K., Kanai, R., & **Baltieri, M.*** (2025). Prior preferences in active inference agents: soft, hard, and goal shaping. *arXiv preprint arXiv:2512.03293*. Under review.
- **Baltieri, M.***, Torresan, F., & Nakai. (2025). A coalgebraic perspective on predictive processing. *arXiv preprint arXiv:2508.16877*. Under review.
- Rosas, F.*, Boyd, A., & **Baltieri, M.** (2025). AI in a vat: Fundamental limits of efficient world modelling for agent sandboxing and interpretability. In *Reinforcement Learning Conference - RLC 2025*.
- Virgo, N.*, **Baltieri, M.**, Biehl, M., & Capucci, M. (2025). A good regulator theorem for embodied agents. In *Artificial Life Conference Proceedings - ALIFE 2025*.
- **Baltieri, M.***, Biehl, M., Capucci, M., & Virgo, N. (2025). A Bayesian interpretation of the internal model principle. *arXiv preprint arXiv:2503.00511*. Under review.

ADDITIONAL PUBLICATIONS

- **Baltieri, M.***, Rosas, F. E., & Torresan, F. (2026). A Braitenberg vehicle’s habitat. In *ALIFE 2026*.
- Torresan, F., & **Baltieri, M.*** (2024). Disentangled representations for causal cognition. *Physics of Life Reviews*.
- **Baltieri, M.***, Iizuka, H., Witkowski, O., Sinapayen, L., & Suzuki, K. (2023). Hybrid Life: Integrating Biological, Artificial, and Cognitive Systems. *WIREs Cognitive Science*. Invited.
- Bruineberg, J., Dolega, K., Dewhurst, J., & **Baltieri, M.*** (2022). The Emperor is Naked: Replies to the commentaries on the target article. *Behavioral and Brain Sciences*.
- Bruineberg, J., Dolega, K., Dewhurst, J., & **Baltieri, M.*** (2021). The Emperor’s New Markov Blankets. *Behavioral and Brain Sciences*.
- Mannella, F., Maggiore, F., **Baltieri, M.**, & Pezzulo, G.* (2021). Active inference through whiskers. *Neural Networks*.
- Hipólito, I., **Baltieri, M.**, Friston, K., & Ramstead, M. J.* (2021). Embodied skillful performance: Where the action is. *Synthese*.
- **Baltieri, M.***, Buckley, C. L., & Bruineberg, J. (2020). Predictions in the eye of the beholder: an active inference account of Watt governors. In *ALIFE 2020*.
- **Baltieri, M.*** (2020). A Bayesian perspective on classical control. In *IJCNN 2020*. IEEE.
- Tschantz, A., **Baltieri, M.**, Seth, A., & Buckley, C. L. (2020). Scaling Active Inference. In *IJCNN 2020*. IEEE.
- **Baltieri, M.***, & Buckley, C. L. (2019). PID control as a process of active inference with linear generative models. *Entropy*, 21(3), 257.
- **Baltieri, M.***, & Buckley, C. L. (2019). Active inference models of motor control without efference copy. In *CCN 2019*.
- **Baltieri, M.***, & Buckley, C. L. (2019). The dark room problem in predictive processing and active inference, a legacy of cognitivism? In *Artificial Life Conference Proceedings - ALIFE 2019*.
- **Baltieri, M.***, & Buckley, C. L. (2019). Nonmodular architectures of cognitive systems based on active inference. In *IJCNN 2019*. IEEE.
- **Baltieri, M.***, & Buckley, C. L. (2018). A probabilistic interpretation of PID control. In *SAB 2018*. Springer.
- **Baltieri, M.***, & Buckley, C. L. (2018). The modularity of action and perception revisited using control theory and active inference. In *ALIFE 2018*.
- **Baltieri, M.***, & Buckley, C. L. (2017). An active inference implementation of phototaxis. In *ECAL 2017*.

SELECTED INVITED TALKS

- *Proving AI safety - Mathematical foundations of AI safety*, AIRJ Symposium 2025: Building Trustworthy AI and Advancing Collaboration Between Italy and Japan, Embassy of Italy, Tokyo, JP Nov 2025
- *Life and its relatives, Breakthrough Initiatives*, Oxford, UK Apr 2025
- *When does a system model another system?*, Internet of Brains Event, Tokyo, JP Mar 2025
- *Foundations of AI: building reliable, trustworthy and agentic systems*, FUTURA @ Italian Chamber of Commerce in Japan, Tokyo, JP Oct 2024
- *Using goals to compress models: effects of bisimulations for (PO)MDPs on embodied agents*, Theoretical and Experimental Approaches to Goal-Directed Behaviour, BCAM, Bilbao, ES Oct 2024
- *LLMs beyond Transformers* (panel discussion), Tokyo AI Talks @ GHOCV, Tokyo, JP Jan 2024

TEACHING

Lecturer in Mathematical Foundations of Agency, CHAIN, JP Dec 2022 - Feb 2023
Lecturer in Machine Learning, University of Sussex, UK Feb 2019 - Jun 2019

EDUCATION

Ph.D. in Informatics and Artificial Intelligence, University of Sussex, UK 2019
M.Sc. in Evolutionary and Adaptive Systems, University of Sussex, UK 2014
B.Eng. in Information and Business Organisation, Università di Trento, IT 2012